ME228 Project Progress Report

Data-Driven Modeling of Fatigue Strength in Low-Alloy Steels

Team Members:
Nishit Dharamshi (24b2153) Shourya Saxena (24b2222)
Kapidhwaj Soni (24b2214) Rudra Khandelwal (24b)

Dataset: NIMS Steel Fatigue Dataset
Reference: Agrawal et al. (2014)

1 Problem Understanding

The Problem: Fatigue accounts for 80–90% of mechanical structural failures. Because physical fatigue testing is slow and expensive, our goal is to predict the rotating bending fatigue strength (MPa) of low-alloy steels using only their chemical composition and heat treatment parameters.

Current Pivot: We shifted from training a single global model to building regime-specific sub-models. This was necessary to handle the bimodal nature of the data, as carburized and non-carburized steels have fundamentally different failure mechanics.

2 Approach

- **Model Selection:** We matched algorithms to the physical data regimes. We use XGBoost for standard through-hardened steels (which form the bulk of our data) and a Random Forest for the smaller, highly specific carburized population.
- **Feature Reduction:** To prevent overfitting and ensure the model remains interpretable for materials engineering, we reduced our features using a two-stage pipeline.
  - *VIF Filtering:* Stripped out highly collinear processing variables (e.g., redundant tempering temperatures).
  - *SHAP Analysis:* Kept only the features with the highest marginal impact on fatigue strength.
- **Reliability Integration:** Instead of just outputting a point prediction, we built a function that maps the model's RMSE against an applied operational load to calculate the component's actual Probability of Failure.

3 Data Status

- **Dataset Details:** The NIMS dataset contains 437 experimental records and 26 original features.
- **Engineered Flags:** We created binary flags (`is_carburized`, `is_through_hardened`) to clearly separate the metallurgical processes.

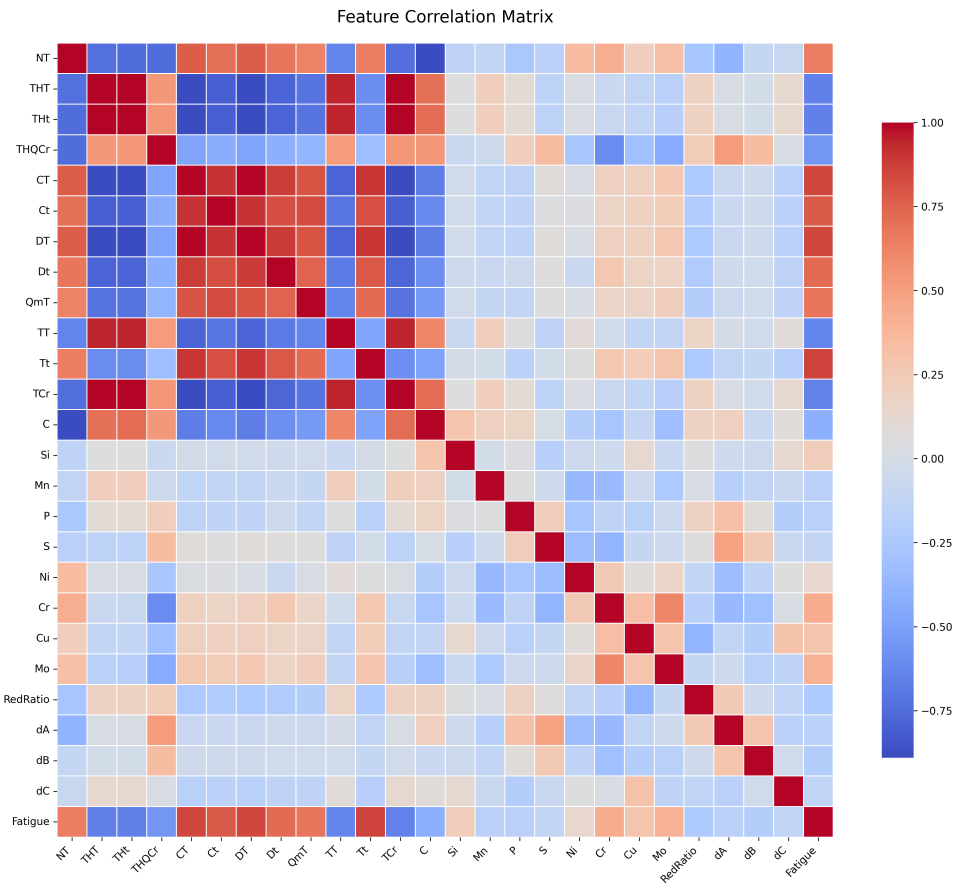

Figure 1: Feature Correlation Matrix demonstrating collinearity among variables.

- **Stratified Splitting:** We implemented stratified splits to ensure the rare carburized samples (∼11% of the dataset) are correctly distributed across our training and testing sets.
- **Strict Scaling:** Features were scaled strictly on the training folds during linear baseline testing and feature selection to prevent data leakage.

4 Initial Experiments

Baseline Limitations: Initial runs using Linear Regression, Ridge, and Lasso proved that purely linear models cannot capture the complex interactions between alloying elements and heat treatments.

> **Key Takeaway:** Splitting the dataset by physical manufacturing process drastically improved our error metrics, confirming that our initial global models struggled by trying to average out two completely different metallurgical regimes.

Sub-Model Performance:

5 Progress & Next Steps

- **Hyperparameter Tuning:** We plan to use Nested Cross-Validation and Optuna to fine-tune tree depth and learning rates, specifically focusing on preventing overfitting in the

Metallurgical Group	Algorithm	RMSE (MPa)	R^2 Score	Feature Count
Non-Carburized	XGBoost	23.29	0.9518	12
Carburized	Random Forest	27.19	0.8522	Reduced

Table 1: Performance metrics of regime-specific sub-models.

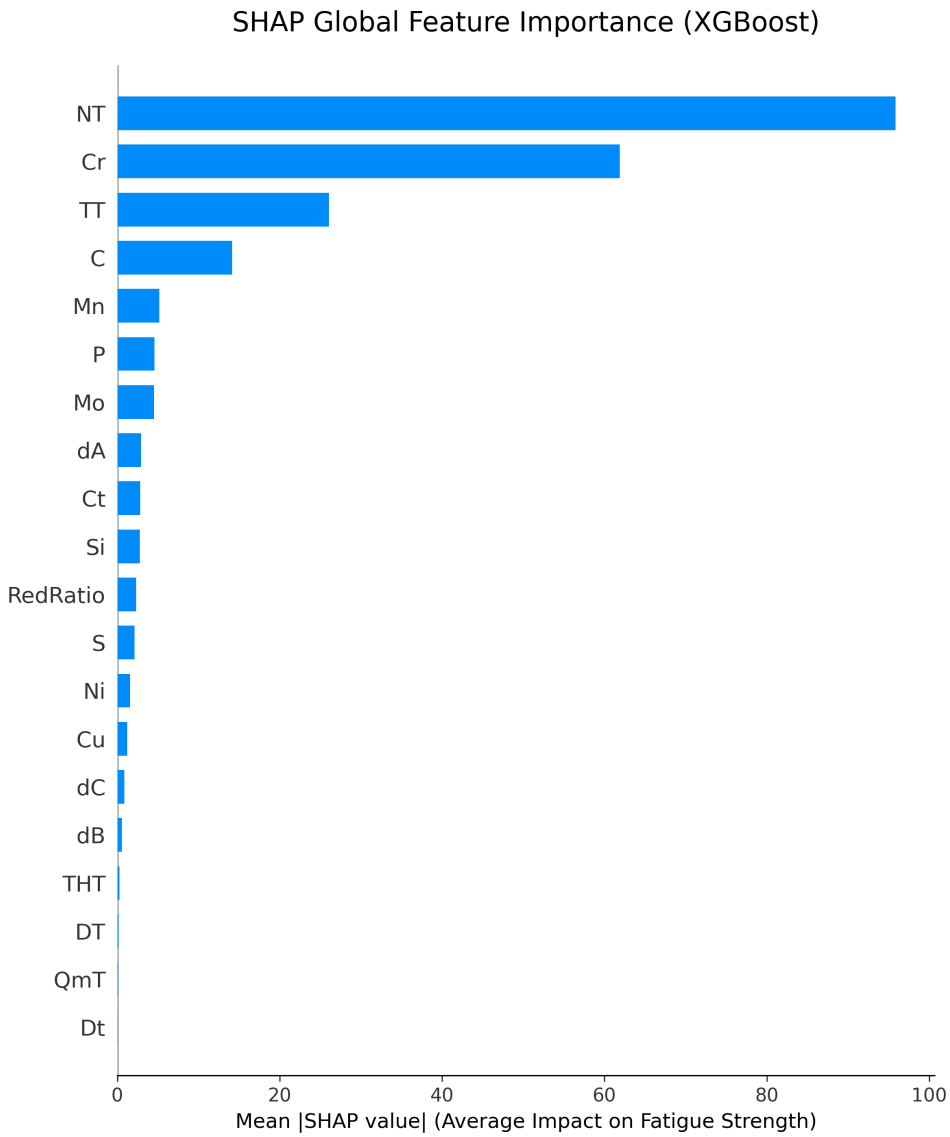

Figure 2: SHAP Global Feature Importance highlighting the most impactful variables.

small carburized dataset.

- **Alloy Recommender (Inverse Design):** We are building a reverse-lookup system. An engineer inputs a target Factor of Safety and operational stress, and the model recommends viable alloy configurations.
- **Cost Analysis Integration:** We aim to add a secondary scoring function to rank these alloy recommendations based on the economic cost of the required alloying elements (Ni, Cr, Mo).

- **Deployment:** If time permits, we will wrap the final models and cost-analysis tool into a Streamlit GUI for real-time engineering use.